STRUCTURE-FIRST-ML

Collected Technical Reports

Structure-first machine learning for physical data

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Preface. Most machine-learning pipelines coerce data into the shape the model expects: irregular observations are interpolated onto a regular grid, point clouds are voxelised, curved domains are flattened, and conserved quantities are left to be learned by accident. Each step discards structure that was present in the measurement and introduces artefacts that were not. The work collected here reverses that order of reasoning — identify the mathematical object the data belongs to, choose a representation with the invariances that object actually has, and only then attach a learning algorithm. Where the representation carries a theorem, the theorem is stated and its hypotheses are tested against real data rather than assumed. Results are reported in both directions: the cases where the approach wins and the cases where it loses are given equal space, because a representation that is claimed to help everywhere has not been understood anywhere.

Data courtesy of the Zwicky Transient Facility, the ZTF Bright Transient Survey, the ALeRCE broker, the Global Meteor Network (CC BY 4.0) and the IAU Meteor Data Center.

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Chapter 1

Path signatures for irregularly sampled light curves

projects/t1-signature-lightcurves

Abstract

A photometric light curve is a finite sample of a path, not a vector: observations arrive at times set by weather and scheduling, in whichever filter was in use, with per-point uncertainties varying by an order of magnitude. Standard pipelines destroy that structure at the first step by interpolating onto a regular grid. The signature transform of rough path theory consumes the raw stream directly and is invariant under reparameterisation of time. This chapter tests whether that theoretical advantage survives contact with real survey photometry, on 2,375 spectroscopically classified ZTF transients in eight classes. The answer is qualified, and the qualifications are the substance of the chapter.

1.1 Introduction

The signature of a path has been applied once in astronomy, to radio-frequency interference detection in interferometric visibility streams [1]. Time-domain photometric classification is untouched. That gap is narrow enough to be worth attempting and broad enough to be worth reporting either way.

The incumbent methods are not weak. Hand-crafted variability features in the lineage of FATS and `feets` have absorbed two decades of domain knowledge; Gaussian-process interpolation is the principled treatment of irregular sampling; random convolutional kernel methods such as MiniRocket [4] are cheap, deterministic and strong. A new representation earns its place only by beating these or by adding something they lack.

1.2 Mathematical background

Definition 1 (Signature)**.** *For a continuous path $X : [0, T] \to \mathbb{R}^d$ of bounded variation, the signature is the sequence of iterated integrals*

$$\mathbf{S}(X) = \left(1,\ \int_{0<t<T} \mathrm{d}X_t,\ \iint_{0<s<t<T} \mathrm{d}X_s \otimes \mathrm{d}X_t,\ \dots\right),$$

whose k-th term is an element of $(\mathbb{R}^d)^{\otimes k}$.

Three properties motivate its use here.

Reparameterisation invariance. For any increasing reparameterisation φ, $\mathbf{S}(X \circ \varphi) = \mathbf{S}(X)$. The signature encodes the order and shape of what happened, not the clock on which it happened. When observation times are set by the weather rather than by the source, this is the correct invariance to have; interpolation-based pipelines must learn it from data instead.

Faithfulness. The signature determines the path up to tree-like equivalence, informally up to retracing [2]. Nothing is discarded but the parameterisation.

Universality. Linear functionals of the signature approximate continuous functions on path space uniformly on compact sets. A linear model on signature features is therefore a universal approximator on paths, which converts architecture design into a question of truncation depth.

For a piecewise-linear path the signature has a closed form. Each straight segment with increment δ contributes the tensor exponential $\delta^{\otimes k}/k!$ at level k, and segments concatenate by Chen's identity

$$\mathbf{S}(X * Y)_k \;=\; \sum_{i=0}^{k} \mathbf{S}(X)_i \otimes \mathbf{S}(Y)_{k-i}.$$

Truncating at depth m retains $\sum_{k=1}^{m} d^k$ coefficients.

1.2.1 Implementation and verification

The transform is implemented from scratch rather than imported, because the theory track requires access to individual tensor levels and because the standard C++ backends do not build reliably on current Python versions. Correctness is established two ways: agreement with `esig`, `signax` and `roughpy` to $\sim 10^{-13}$ across four path configurations, and four structural identities checked directly — reparameterisation invariance, level one equalling the total increment, the shuffle identity $S_1 S_2 = S_{12} + S_{21}$, and the vanishing of the signature of a path concatenated with its own reversal. All sixteen checks pass.

1.3 Where the ordering information lives

Before applying the transform to real data it is worth confirming that the claimed mechanism exists, in a setting where the answer follows from the construction. Two classes of synthetic double-peaked light curve are built to differ only in which peak comes first: the peaks sit symmetrically about the midpoint of the window and the curve starts and ends at the same baseline, so one class is the exact time reverse of the other. The marginal distribution of magnitudes is therefore identical between classes, and so is the net change from first to last observation.

The prediction recorded in advance was that separation would appear at depth 2, the Lévy area being the classical order-sensitive term. It does not.

The reason is geometric and specific to this path construction. With channels (t, m) the time coordinate is strictly increasing, so the antisymmetric part of level 2 — the Lévy area —

Representation	Balanced accuracy
Order-blind summary features	0.512
Signature depth 1	0.524
Signature depth 2	0.489
Signature depth 3	**0.978**
Signature depth 4	0.990

Table 1.1: Separation of two synthetic classes differing only in the order of two peaks. Chance is 0.5. Ordering becomes visible at level 3, not level 2.

reduces to $\int m\,\mathrm{d}t$ once the boundary terms vanish. That is the area under the light curve, which is the same whichever peak came first. A path monotone in one coordinate encloses no signed area that ordering can change. Ordering first becomes visible at level 3, in terms of the form $\iiint \mathrm{d}t\,\mathrm{d}m\,\mathrm{d}t$, which weight magnitude excursions by when they occurred.

The practical consequence is direct: truncating at depth 2 to economise on features would discard precisely the information the method exists to supply, and the resulting null result would have been attributed to the astrophysics.

1.4 Data

Labels are the spectroscopic classifications of the ZTF Bright Transient Survey; per-epoch photometry in g and r is served by the ALeRCE broker. ALeRCE's own machine classifications are never used, since training on one classifier's output and reporting agreement with it measures nothing.

After quality cuts requiring at least ten detections and at least three in each band, the sample is 2,375 objects with a median of 42 detections spanning a median of 122 days, in eight classes: CV (456), SN II (426), SN Ia (406), AGN (376), SN IIn (273), SN Ic (185), SN Ib (148) and SLSN-I (105).

A first version of the fetch submitted objects grouped by class while the broker's failure rate rose during the run, so classes submitted last lost far more objects than those submitted first and SN Ib was eliminated entirely. That is a selection effect on the label rather than random attrition, and it would have corrupted every downstream number. After shuffling before any network call, lowering concurrency and adding exponential backoff, the per-class retrieval rate is 1.0 across all eight classes.

1.5 Method

1.5.1 Constructing the path

The difficult part is not the signature but deciding what the path is. Observations in different filters are not simultaneous, so no instant exists at which a two-band vector is defined, and any joint multi-channel construction embeds a choice. Three are implemented and compared rather than assumed: *per-band*, in which each filter is its own two-dimensional path and the signatures are concatenated, with no cross-band information and no interpolation; *interleaved*, in which all observations in time order form one path with a band indicator channel; and *joint forward-fill*, a common time axis with the last observed magnitude carried forward, which is piecewise-constant interpolation and is labelled as such.

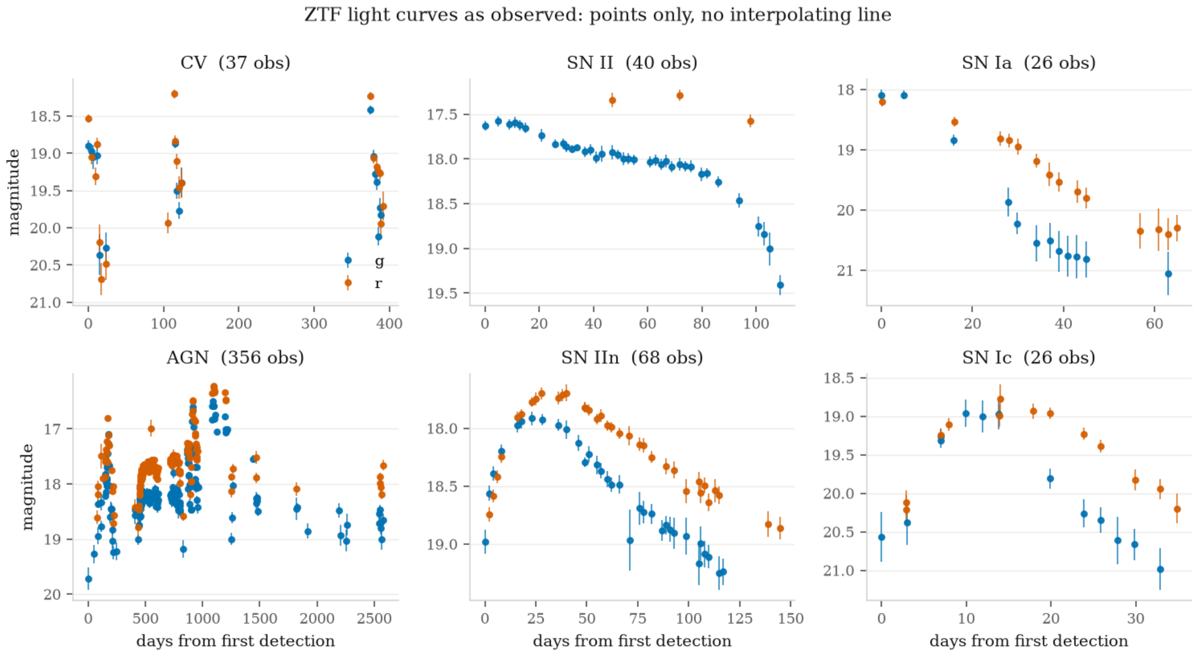

Figure 1.1: Example light curves, one per class, drawn as points with error bars and no connecting line: a line between observations would assert a continuity the data does not contain. Note the range of durations, from 26 detections over 65 days to 356 over 2,500.

1.5.2 Channel preparation

How the channels are prepared turned out to matter more than which path construction was used. Scaling time to $[0, 1]$ discards the duration of the event; centring magnitudes discards absolute brightness. Both are strong physical discriminants for transients, and the hand-crafted baseline retains them as `t_span` and `peak_mag`. Discarding them and then reporting that signatures lose would measure the preprocessing rather than the representation.

A related fact is worth stating because it is a check on the implementation rather than a result: the *raw* and *raw time* preparations score identically to four decimal places. The signature depends only on increments, so a constant magnitude offset cannot change it. Absolute brightness can enter only through a basepoint, and when introduced that way it degrades performance rather than improving it.

1.5.3 Baselines

Every representation is evaluated on identical cross-validation folds with identical classifiers. The hand-crafted baseline implements the FATS and `feets` lineage from scratch; almost all of its features are functions of the marginal magnitude distribution and are therefore invariant to permuting observations in time, which is exactly the information signatures retain. The three ordering-sensitive features it does contain — linear trend, rise and fade fractions, maximum slope — are included deliberately, since omitting them would make the comparison dishonest. MiniRocket is given linearly interpolated fixed-length input, which is its best case; that interpolation is precisely the cost the signature approach claims to avoid, and granting the baseline its best case is the only way to measure that cost.

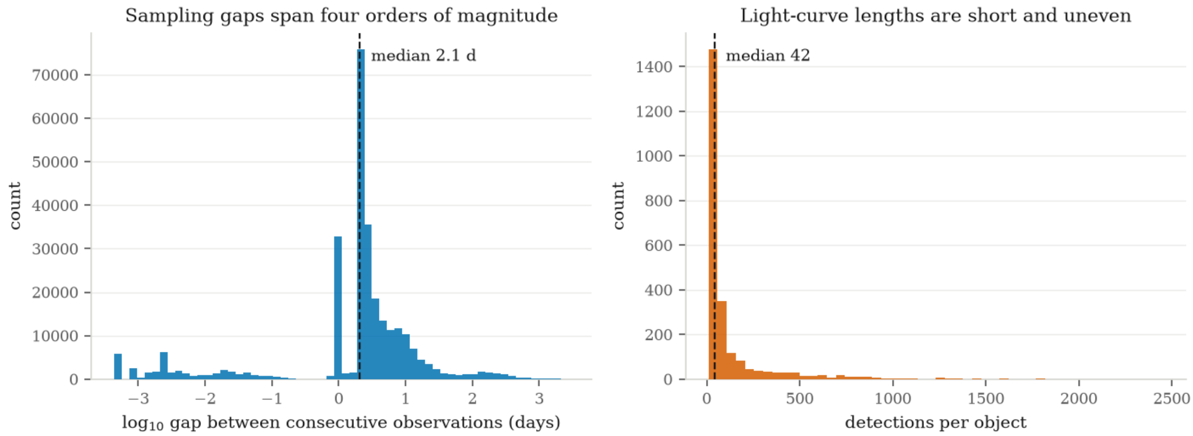

Figure 1.2: The sampling gap distribution is bimodal and spans four orders of magnitude: repeated exposures within a single night near 10^{-3} days, the main survey cadence at 1 to 10 days, and seasonal visibility windows at 100 to 1,000 days. Same-night exposures occupy almost no arc length and so barely move the signature, whereas a fixed-grid method weights them like any other sample.

1.6 Results

All figures are balanced accuracy under stratified cross-validation with gradient-boosted trees, on 2,375 objects in eight classes. Chance is 0.125.

1.6.1 Preprocessing dominated the representation

Representation	Features	Balanced accuracy
summary + signature (lead-lag, depth 4)	729	**0.578**
summary (hand-crafted)	49	0.560
signature, per-band, raw time, lead-lag, depth 4	680	0.548
MiniRocket (interpolated grid)	9,996	0.538
signature, per-band, raw time, depth 4	60	0.524
signature, per-band, unit time, depth 4	60	0.455

Table 1.2: Head-to-head comparison. The gap between the last two rows is the cost of discarding event duration by scaling time to $[0, 1]$; the gap between rows three and five is the gain from the lead-lag transform.

Scaling time to $[0, 1]$ costs 0.069, because it discards the duration of the event, which the hand-crafted baseline retains as `t_span`. The first version of this pipeline did exactly that, and would have reported a far worse number as a property of signatures rather than of the preprocessing. The lead-lag transform is worth a further 0.025, taking signatures past MiniRocket but not past the incumbent; it exposes quadratic variation, and roughly a third of the sample (AGN and CV) is stochastically variable rather than transient.

A third observation is a check on the implementation rather than a result: the *raw* and *raw time* preparations score identically to four decimal places. The signature depends only on increments, so a constant magnitude offset cannot change it. Absolute brightness can enter only through a basepoint, and introduced that way it degrades performance rather than improving it.

1.6.2 Complementarity, with a matched-noise control

Signatures alone lose to a 49-feature baseline while using fourteen times as many features. The concatenation scores above the baseline, but by 0.015 against a fold spread of 0.020, which on its own establishes nothing. The question was therefore given a dedicated test: fifty paired folds on identical splits, a Wilcoxon signed-rank test, a bootstrap interval, and a control in which the signature block is replaced by Gaussian noise of identical shape and column variance.

Feature block	Balanced accuracy
summary + signature	**0.5769 ± 0.0200**
summary	0.5623 ± 0.0167
signature alone	0.5473 ± 0.0218
summary + matched noise	0.5278 ± 0.0146

vs. summary	Difference	Folds improved
+ signature	+0.0147	38 / 50
+ matched noise	−0.0345	1 / 50

Table 1.3: Left: mean and standard deviation across fifty paired folds. Right: paired differences against the baseline, with 95% bootstrap intervals $[+0.0078, +0.0212]$ and $[-0.0396, -0.0297]$ respectively.

The control is what makes the result readable. A gradient-boosted ensemble handed 680 extra candidate splits might improve for reasons unrelated to their contents; it does not. Six hundred and eighty columns of noise *cost* 0.0345, while the same number of signature columns *gain* 0.0147, under identical folds and identical dependence structure.

One caveat belongs beside the p-value (9.5×10^{-5}): repeated cross-validation folds reuse the same 2,375 objects and are not independent, so both it and the bootstrap interval are optimistic as formal inference. The comparison that does not rest on that assumption is the one against the noise control, and it is the one the conclusion leans on.

1.6.3 The sampling ablation

The headline accuracy on full light curves says little, because every representation does reasonably well when the data are dense. The claim under test concerns *where* signatures should help, so the light curves were thinned to fixed retained fractions under two regimes removing the same number of observations while destroying different structure: *random* thinning, with each observation dropped independently, and *blocked* thinning, with contiguous runs removed to imitate weather outages and seasonal windows.

	random				blocked		
Representation	1.00	0.60	0.35	0.20	0.60	0.35	0.20
summary	0.560	0.549	0.532	0.513	0.536	0.520	0.498
signature	0.538	0.519	0.497	0.503	0.508	0.487	0.462
MiniRocket	0.538	0.504	0.478	0.467	0.481	0.431	**0.398**

Table 1.4: Balanced accuracy against retained fraction under both thinning regimes.

The informative quantity is the difference between regimes at matched density — the cost of gap *structure* with gap *count* held fixed:

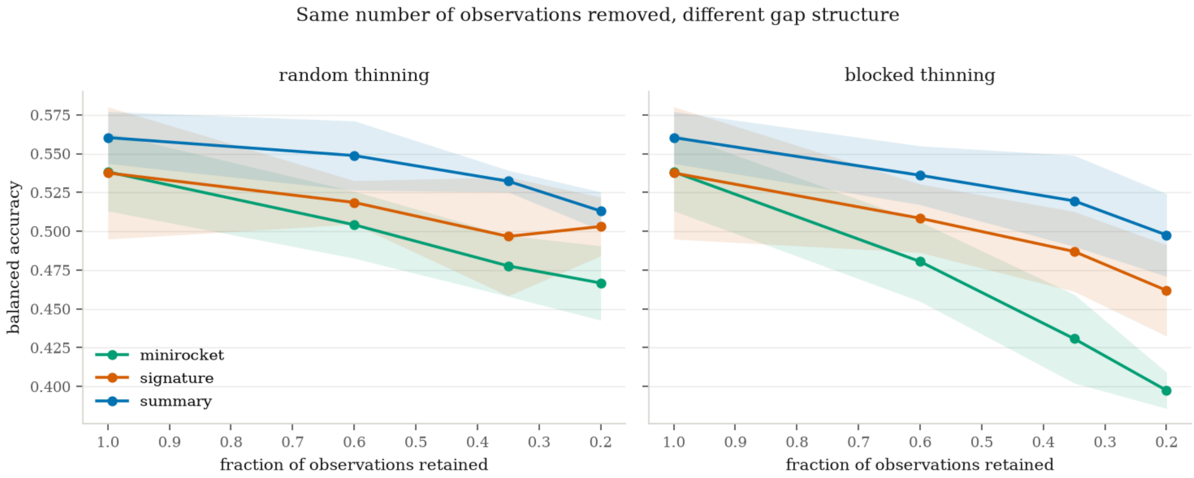

Figure 1.3: The same number of observations removed under two gap structures. MiniRocket, which requires a regular grid and is given one, collapses under blocked thinning while the other two degrade gently.

Retained fraction	summary	signature	MiniRocket
0.60	−0.013	**−0.010**	−0.024
0.35	−0.013	**−0.010**	−0.047
0.20	**−0.016**	−0.041	−0.069

Table 1.5: Structured-gap penalty: blocked minus random at equal retained fraction.

Gap structure matters, not merely gap count, and it costs the interpolation-dependent method the most. MiniRocket pays two to four times the penalty of the other two at every density, losing 0.069 to structured gaps alone at 20% retention. This is the strongest evidence in the chapter for the premise that resampling irregular data onto a grid discards information, and it is a prediction that could have failed: identical curves under the two regimes would have collapsed the motivation entirely.

At moderate thinning the signature is the least damaged by gap structure, which is the direction the reparameterisation argument predicts. At the sparsest setting it is not: its own structured-gap penalty jumps to −0.041, worse than the baseline's −0.016. Below roughly a dozen observations per object the path is too coarsely sampled for the iterated integrals to be estimated stably, and the invariance argument stops paying.

1.6.4 The Gaia counter-test

The counter-test was registered with the main experiment and run afterwards on 2,999 Gaia DR3 periodic variables in six classes. Its logic is that ZTF transients are separated by the shape and ordering of the light curve, which signatures encode, whereas Gaia periodic variables are separated by *period*, which is a statement about the clock — and the signature is invariant to reparameterisation of the clock.

Prediction P1 is confirmed: signatures lose six times harder on periodic variables, with a label-shuffled control at 0.160 against a chance value of 0.167.

Prediction P2 is refuted. Restoring duration was expected to matter more on Gaia and matters less. The cause is measurable: Gaia's observing baseline is set by the mission rather

Measurement	Gaia (2,999)	ZTF (2,375)
Baseline minus best signature	**+0.0743**	+0.0123
Value of keeping duration	+0.0387	**+0.0688**
Value of the lead-lag transform	**+0.1130**	+0.0246
Complementarity: signature added to baseline	**−0.0000**	**+0.0147**
Matched-noise control	−0.0016	−0.0345

Table 1.6: All figures on full samples. An earlier 300-object pilot overstated the central gap by more than a factor of two, which is why pilot numbers were never carried into a claim.

than by the object, so span carries almost no information (coefficient of variation 0.039 against ZTF's 1.250). **Duration is not period.** The time channel restores how long the observation lasted, not the clock the variability lives on.

The sharpest form of the argument is a single feature. Restricted to the 1,806 objects carrying a catalogued period:

Arm	Features	Balanced accuracy
period only	**1**	$\mathbf{0.7726 \pm 0.0925}$
signature, per-band, raw time, lead-lag, depth 4	1,020	0.7412 ± 0.0128
summary (hand-crafted)	72	0.9850 ± 0.0069
summary + period	73	$\mathbf{0.9990 \pm 0.0013}$

Table 1.7: One number outperforms a 1,020-coefficient signature representation. The period requirement retains classes unevenly (ECL 100%, CEP 98%, RR 98%, LPV 45%, DSCT-group 19%, RS 1.2%), so this arm speaks about the periodic classes rather than the whole sample.

Complementarity also vanishes. On ZTF, signatures added +0.0147 against a noise control of −0.0345; on Gaia the same test returns −0.0000. Where the baseline already reaches 0.985, signatures contribute nothing.

1.6.5 What the prediction got right and wrong

The prediction registered before any data was touched had two halves, and the measurements split them cleanly.

Confirmed. Gap structure matters independently of gap count, and the cost falls hardest on the method that requires a regular grid. That is the premise the whole track rests on, and it was falsifiable: identical degradation curves under random and blocked thinning would have ended the argument.

Not confirmed. Signatures do not close the gap as data thins. The baseline-to-signature difference under random thinning runs $0.023 \to 0.030 \to 0.036 \to 0.010$ as the retained fraction falls from 1.00 to 0.20 — widening through the middle of the range rather than shrinking monotonically — and at the sparsest setting the signature becomes *more* sensitive to gap structure than the baseline, not less.

What survives. Signatures alone do not beat mature hand-crafted features at any sampling density tested. They do carry information those features lack, established against a matched-noise control. Not a better representation — a different one, whose advantage is over grid-based methods rather than over domain knowledge.

And the counter-test closes it symmetrically. The property that produces the win produces the loss. Signatures carry information hand-crafted features lack when classes differ by the shape and ordering of an irregularly sampled curve, and carry nothing when classes differ by a period the representation is constructed to ignore — where a single catalogued number outperforms a thousand signature coefficients. A representation that helped everywhere for the same stated reason would not have been understood; measuring where it fails is what makes the account testable.

Chapter 2

Consensus clustering for meteoroid streams

projects/t2-meteor-stream-discovery

Abstract

Meteoroid stream identification is a clustering problem in which the metric is dictated by celestial mechanics rather than convenience, and the classical orbital dissimilarity criteria disagree with one another in ways that propagate into which streams get found. This chapter implements the procedure the field's own 2025 methods review recommends and nobody had run on the Global Meteor Network archive: cluster under several criteria at once and keep only what survives all of them. Applied to 2,146,868 orbits, it recovers 57% of the established IAU catalogue and proposes nothing new. The substantive result is not the null on discovery but a measured demonstration that the permutation null used widely in threshold-based work cannot separate a stream from sporadic structure at all.

2.1 The archive and the criteria

The GMN publishes monthly trajectory summaries under CC BY 4.0: 86 columns per multi-station meteor, with orbital elements, radiant, geocentric velocity, per-quantity uncertainties, convergence angle and the pipeline's own IAU shower association. After quality cuts on convergence angle, radiant and velocity uncertainty, 2,146,868 of 3,178,785 meteors remain, of which 631,065 carry a shower code across 426 distinct showers. The GMN association is used only as a control, never as an input.

Four criteria are implemented: D_{SH} (Southworth & Hawkins 1963), D_D (Drummond 1981), D_H (Jopek 1993) and D_N (Valsecchi, Jopek & Froëschlé 1999). Verification uses analytic special cases where each formula collapses to closed form — orbits differing only in eccentricity, only in perihelion distance, only in inclination, coplanar orbits differing only in argument of perihelion — plus identity, symmetry and a physical check that a Geminid pair falls below the conventional association threshold while a Geminid–Perseid pair falls far above it. All 23 checks pass.

One correction inside the verification is worth recording. The closed form $D_D = \Delta i/\pi$ for orbits differing only in inclination holds only with the perihelion on the line of nodes; otherwise tilting the plane moves the perihelion direction itself. The first version of the test used a generic perihelion, flagged the implementation as wrong, and a hand computation showed the implementation was right and the expectation was not.

2.2 Why the null model is the whole problem

Pairwise distance over 2.1 million meteors is 4.6×10^{12} pairs, so the sweep runs in overlapping windows of solar longitude — physically motivated, since a stream is active over a bounded range and meteors months apart cannot belong to the same encounter. Within each window, clustering under each criterion is calibrated to a common false-positive rate against a null, and only structures placed together by every criterion are retained.

The first complete sweep returned 1,249 structures of which 498 matched no IAU list, against 64 recovered established showers. A novel-detection rate seven times the known-recovery rate is a method reporting its own false positives. The permutation null — shuffling each orbital element independently — was the leading suspect, and was then measured rather than assumed.

A null has two jobs, pulling in opposite directions. In a region containing only sporadic meteors it must predict roughly the number observed (calibration); in a region containing a stream it must predict far fewer (discrimination). Both are measured as the ratio $n_{\mathrm{exp}}/n_{\mathrm{obs}}$ inside a D_{SH} ball, with regions centred on GMN's own sporadic and shower labels.

Null model	Sporadic ratio	Stream ratio	Separation
sideband — real orbits, adjacent $\lambda_\odot$, nodes rotated	**0.800**	0.107	**7.48**
kde — joint density, bandwidth above the stream scale	0.369	**0.072**	5.12
permutation — marginals only	0.366	0.458	**0.80**

Table 2.1: Median across the Perseid, Southern Taurid and Geminid windows. A separation below one means the null predicts the same density at a real shower as in empty background.

The permutation null scores below one. It cannot distinguish a stream from ordinary sporadic structure, and the reason is structural: the helion, antihelion, apex and toroidal sources exist *because* orbital elements are correlated, so shuffling them independently models a background that does not exist and every real correlation registers as an excess.

Two replacements were implemented so the choice would be measured rather than argued. The KDE null fits the joint distribution with a bandwidth chosen physically — above the internal scale of a stream, below the extent of a sporadic source — and deliberately not cross-validated, since a fitted bandwidth would model the streams too. The sideband null borrows real orbits from solar longitudes either side of the window, past a gap wider than a typical shower's activity. That the two agree at the streams (0.107 and 0.072) is the check on both.

The sideband null required one correction that failed in a way resembling success. Without rotating the borrowed nodes it predicted zero density at sporadic regions *and* zero at the Perseids, which reads as perfect discrimination and is total collapse. A meteoroid is only observable where its orbit crosses Earth's, so its ascending node is locked to the solar longitude of the encounter; orbits borrowed from 18° away carry nodes 18° away, outside every ball centred in the window.

2.3 What the corrected sweep found

The recurrence filter only became a filter afterwards. Requiring that meteors be *present* near the candidate orbit in three or more years separates known from unknown by 2.3 percentage points

	Permutation null	Sideband null
Distinct structures	1,249	358
Unmatched to any IAU list	498	**56**
Surviving density excess in ≥ 3 apparitions	461	**21**
Established showers recovered	64 of 113	64 of 113

Table 2.2: Replacing the null cut unmatched structures 8.9-fold with no loss of recovery.

— it measures nothing, because the sporadic background is present every year too. Requiring a density *excess* under the permutation null gives 4.5 points. The same statistic under the sideband null gives 78.2% against 37.5%, a separation of 40.7 points. With a null that cannot see streams, no statistic built on it can either.

Of the 21 survivors, 8 have a median per-year $z \geq 3$ across at least five apparitions. Rechecked against every IAU entry at loose tolerance, **eight of eight are known showers**, two of them within 2°: ϵ-Draconids at 1.3° and Daytime μ-Cancrids at 1.8°. The strict tolerance rejected them because catalogue V_g values come from different solutions than the one measured here. The remaining false positives therefore arise in the *matching* step, not the clustering: the clustering finds real streams and the bookkeeping failed to recognise them.

2.4 Conclusion

No new stream is claimed. What stands is a verified implementation, a null result on discovery at 57% completeness against the established catalogue, and a quantitative comparison of sporadic-background models showing that the permutation null cannot do the job it is routinely asked to do while a sideband null of real orbits with rotated nodes can. That last result is independent of whether anything new is found, and it is the part worth carrying forward.

Chapter 3

Where ordering lives in the signature

projects/t3-signature-theory

Abstract

The theory track exists because Chapter 1 produces numbers that do not explain themselves. Two questions were asked and both answers contradicted the prediction registered for them. Ordering information appears at truncation depth three rather than depth two, for a reason that holds for every time-augmented signature of a time series and not merely for the construction tested. And the lead-lag transform, the obvious candidate for repairing that, does not repair it — the refutation is exact rather than statistical. A different family of augmentations does work, and the mechanism is identifiable.

3.1 The degeneracy at level two

Two classes of synthetic double-peaked curve were built to differ *only* in which peak comes first: identical marginal magnitude distributions, identical net change from first to last observation, opposite ordering. Any order-blind representation must fail on them by construction.

Representation	Balanced accuracy
Order-blind summary features	0.512
Signature depth 1	0.524
Signature depth 2	0.489
Signature depth 3	**0.978**
Signature depth 4	0.990

Table 3.1: Chance is 0.5. Depths 1 and 2 sit at it; separation appears at depth 3.

The prediction was depth 2, the Lévy area being the classical order-sensitive term. The reason it fails is geometric. With channels (t, m) the time coordinate is strictly increasing, so the antisymmetric part of level 2 reduces to $\int m \, \mathrm{d}t$ once the boundary terms vanish for a path returning to its starting magnitude. That is the area under the curve, identical whichever peak came first. **A path monotone in one coordinate encloses no signed area that ordering can change.**

This generalises: every time-augmented signature of a time series has a monotone time channel, because that is what time augmentation means. Truncating at depth 2 to economise on features therefore discards precisely the ordering information the method exists to supply, and the resulting null result would be misattributed to the data.

3.2 Lead-lag does not repair it

The lead-lag transform doubles the channels and breaks monotonicity, so it was the obvious candidate — and Chapter 1 had measured it as worth +0.025 on real photometry without knowing why. On exact time-reversed pairs, the maximum absolute difference between the two classes' signature tensors:

Augmentation	Level 1	Level 2	Level 3	Level 4
time-augmented	5.6×10^{-17}	2.2×10^{-16}	0.360	0.546
lead-lag + time	5.6×10^{-17}	0.0136	0.277	0.486
lead-lag, magnitude only	0.0	5.1×10^{-16}	0.102	0.147
cumulative integral	5.6×10^{-17}	0.0601	0.360	0.546
running maximum	5.6×10^{-17}	**0.2739**	0.420	0.599

Table 3.2: Lead-lag's level-2 content on magnitude alone is zero to machine precision. Its entire depth-≤ 2 content is quadratic variation, which is invariant under permutation of increments and therefore order-blind by construction.

What does work is a *causal, memory-carrying* channel. A running maximum reaches at level 2 what the plain path only reaches at level 3. These channels are not reparameterisation-invariant and are not meant to be: they encode what had happened by now, which is exactly the ordering content level 2 cannot otherwise see. The result is candidate-level and synthetic; no photometry was touched.

3.3 Making the checks capable of failing

Every verification was deliberately broken to confirm it could fail. The correct level-2 computation gives 8.9×10^{-16}; zeroing the Q_{tm} term gives 0.044, zeroing Q_{tt} and Q_{mm} gives 2.00, transposing the tensor index gives 0.082, and an arbitrary non-symmetric sampling grid gives 0.160. A test that passes while measuring nothing is worse than no test, and this repository had already produced one — a contrast check that returned the same number under three different conditions before the construction was corrected.

Chapter 4

Plate topology: a century of sky as image structure

projects/t4-plate-topology

Abstract

This chapter moves from time series to images, and reuses a method the repository had already rejected. Persistent homology was dismissed for Gaia phase space because degree-1 Vietoris–Rips is intractable on high-dimensional point clouds. On a pixel grid the natural construction is a cubical complex with a sublevel-set filtration, which is near-linear: four million cells in seconds. The same mathematics that was unusable on a point cloud is routine on an image. The premise was tested before any pipeline was built. It survives three experiments and fails the fourth: these diagrams distinguish one patch of sky from another but not a patch of sky from its own past, not even across the eruption of one of the brightest novae on record.

4.1 The archive and the opening

The Harvard plate collection covers the sky from the 1880s to the 1990s — 429,274 plates, digitisation completed in 2024 — and carries what no modern survey has, a century-long time axis on the same patches of sky. Of twenty arXiv papers using it, eighteen extract point-source light curves and the remaining two touch plate imagery only inside the reduction pipeline's own defect handling. Nobody appears to treat a plate series as an (x, y, epoch) object.

Access was verified live with no authentication: 9,036 plate identifiers from one coordinate search, metadata from 1896 onward, 4.8 MB binned mosaics with real WCS, and calibrated 835×835 cutouts at any position.

4.2 A data-product error that would have read as a method failure

The first falsification run used 16×-binned whole-plate mosaics and failed with every patch reporting a bright-pixel fraction of exactly zero. Measured on the same field:

The mosaic is the right product for locating a plate and the wrong one for analysing structure

Product	Background MAD	Fraction above 5 MAD
16×-binned mosaic	833.0	0.00000
Calibrated cutout (1.44″/px)	72.0	0.00522

Table 4.1: Binning collapses stellar profiles below one pixel and folds the vignetting gradient into the scatter, inflating the robust deviation elevenfold.

on it. Unchecked, this would have been recorded as a failure of persistent homology rather than of the product choice.

4.3 Falsification results

On calibrated cutouts, 36 patches from 12 plates. Persistence responds to source content: patches separated twentyfold in measured source fraction differ by a factor of 3.2 in maximum persistence (715 against 2295). But total persistence and the count of significant features are flat or slightly inverted — *one* summary statistic carries the signal, and a careless choice would have missed it or reported the opposite sign.

Against pixel-shuffled controls with identical histograms, real fields give 5,779 components against 7,316 and half the total persistence, with a bottleneck distance 0.347 of the diagrams' own scale.

A registered expectation was wrong here too. The original criterion required real fields to show *longer* lifetimes. Shuffling preserves the histogram, so bright pixels survive but scatter into isolation, and an isolated bright pixel on a flat background is long-lived, while real sources cohere into fewer components. Lifetime direction is not the discriminant; separability is. The criterion was restated and the original expectation left on the record, because a criterion rewritten after seeing the data has to show its working.

4.4 Two gates, and the one that closed the track

The epoch-comparison test in the falsification run came out at 2.2%: within-field bottleneck 1,157 against between-field 1,183. Plate-to-plate variation in emulsion, exposure and limiting magnitude looked comparable to the difference between one patch of sky and another. Measured directly on the plates used, exposure times of 10 s and 300 s give source fractions of 0.003 and 0.218 — a factor of seventy.

4.4.1 Gate one: normalisation, passed, with a refuted argument

Three normalisations were compared on 14 plates spanning 1896–1899, with the threshold fixed at 0.15 before the run.

Normalisation	Within field	Between fields	Relative separation
none	1038.0	1377.0	+0.327
zscore	9.73	15.84	**+0.628**
rank	0.1221	0.1207	**−0.011**

Table 4.2: The rank transform, argued in advance to be the principled choice, came last.

Two things written down in advance were wrong. The module had argued at length that rank normalisation was principled: sublevel-set persistence depends only on the order in which pixels enter the filtration, so rank is the canonical representative of the monotone class. That reasoning applies to the wrong object. Persistence is monotone-invariant in the *shape* of the diagram — which features exist and how they nest. But the diagram's *coordinates* are the filtration values themselves, and bottleneck distance is computed on those coordinates. Rank forces every patch to a uniform distribution on $[0, 1]$, so all diagrams occupy the same narrow range whatever is in them: it preserves the topology and destroys the metric.

Second, the flatness that motivated the gate was a small-sample effect. Unnormalised separation is $+0.327$ here against $+0.022$ in the five-plate run; fourteen plates opened it.

4.4.2 Gate two: temporal change, failed

Separating one field from another is necessary and not sufficient. The track depends on detecting *change* in one field across decades, so the second gate ran on GK Persei — which erupted in February 1901 from magnitude 13 to 0.2 — using 14 plates from 1900–1904 and a control field 0.8° away on the same plates, 91 pairs each.

Field	Median	Mean	Std
Target (contains GK Per)	**1.944**	7.090	9.169
Control	**3.234**	6.348	7.485

Table 4.3: Change signal -0.399 against a threshold of $+0.314$ fixed before the run.

The gate fails, and in the wrong direction: the field containing one of the brightest novae on record is *more* self-consistent across epochs than a field without one. An audit is recorded and does not rescue it — the distributions are extremely skewed, median and mean point opposite ways, and the control contains pairs at exactly zero where cutouts gave identical or empty diagrams. That is a reason to distrust the number rather than to overturn it: the criterion was fixed in advance precisely so it could not be renegotiated afterwards, and nothing in the audit shows the target separating its own epochs better than the control. A method whose verdict flips with the choice of summary statistic is not detecting a thirteen-magnitude nova.

4.4.3 Diagnosis and closure

The failure is dilution, not topology. Gate one's separation is driven by which stars sit in a field, a property that does not change with epoch, and one nova among thousands of sources does not move a whole-field diagram. Localised persistence computed around individual candidate positions might see it, but that is a matched-filter search using topological features rather than the structural-characterisation idea this track was opened to test, and it would need its own premise test.

The track closes. What stands is a verified DASCH access module, a falsification suite that caught a data-product error before it could be mistaken for a method failure, a normalisation comparison that refuted its own principled argument, and a clean negative reached in four experiments rather than a built pipeline.

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